

Grafik yöntemi

$$x_0 = 1.5 \quad \Delta x = 0.75$$

$$f(x) \rightarrow x^2 - 6x + 5$$

-1 -5

$$\epsilon = 0.0235$$

1. iterasyon

<u>x</u>	<u>f(x)</u>
1.5	-1.75
2.25	-3.43
3	-4
3.75	-3.43
4.50	-1.75
5.25	+1.0625

2. iterasyon $\Delta x = 0.75/2 = 0.375$

<u>x</u>	<u>f(x)</u>
4.50	-1.75
4.875	-0.4844
5.25	+1.0625

↓
 Δx

3. iterasyon $\Delta x = 0.375/2 \rightarrow 0.1875$

<u>x</u>	<u>f(x)</u>
4.875	-0.4844
5.0625	0.2639

$$\Delta_x = 0.0625 \quad 0.2551$$

4. iteration $\Delta_x = 0.1875/2 = 0.094$

<u>x</u>	<u>f(x)</u>
4.875	-0.4844
4.969	-0.1230
5.063	0.2559

5. iteration $\Delta_x = 0.094/2 = 0.047$

<u>x</u>	<u>f(x)</u>
4.969	-0.1230
5.016	0.064

Δ_x

6. iteration $\Delta_x = 0.047/2 = 0.0235$

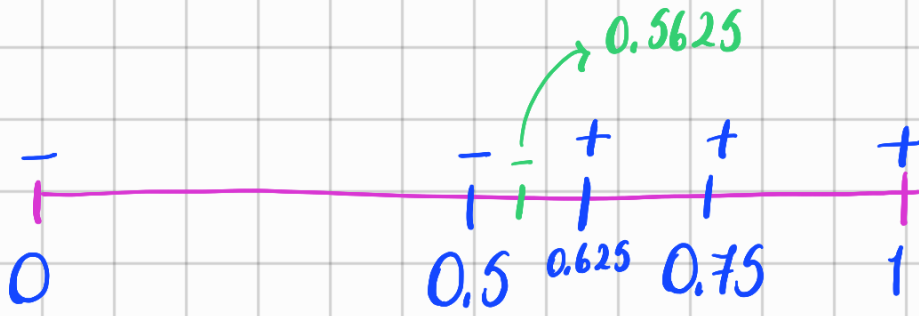
<u>x</u>	<u>f(x)</u>
4.969	-0.1230
4.9923	-0.0299
5.016	0.0640

Δ_x

$\rightarrow f(x) = 0$

$\hookrightarrow |f(x_k)| < \epsilon \quad x_{\text{tök}} = 5$

$$f(x) = x^3 - 7x^2 + 14x - 6 \quad [0,1] \text{ Hata } \epsilon = 0,01$$



1. Adım: $f(a)$ 'yi hesapla
2. Adım: $f(b)$ 'yi hesapla
3. Adım: hata payına bak
4. Adım: Tekrar bir orta nokta belirle
5. Adım: $f(c)$ 'yi bul
6. Adım: $f(c), f(b)$
7. Adım: Hata payına bak

1. iterasyon $[0,1]$

$$f(0) = -6 \quad f(0) + f(1) < 0$$

$$f(1) = 2$$

$$\text{Hata } 1,0 > 0,01 \rightarrow \underline{b-a} < \epsilon$$

$$\text{Hata} = \frac{1-0}{2^1} < 0.01$$

$$C = \frac{1+0}{2} = 0.5$$

$$f(0.5) = -0.625$$

$$f(0.5) \cdot f(1) < 0$$

$$[0.5, 1]$$

$$\text{Hata} \rightarrow \frac{1-0.5}{2^2} < 0.01$$

2. iterasyon

$$f(0.5) \rightarrow -0.625$$

$$f(1) \rightarrow 2$$

$$C = \frac{1+0.5}{2} = 0.75$$

$$f(0.75) = 0.9844$$

$$f(0.5) \cdot f(0.75) < 0$$

3. iterasyon

$$f(0.5) = -0.625$$

$$[0.5, 0.75]$$

$$f(0.75) = 0.9844$$

$$\text{Hata} \rightarrow \frac{0.75 - 0.5}{2^3} < 0.01$$

$$C = \frac{0.75 + 0.5}{2} \rightarrow 0.625$$

$$f(0.625) \rightarrow 0.2598$$

$$f(0.5) f(0.625) < 0$$

4. iterasyon $[0.5, 0.625]$

$$f(0.5) = -0.625$$

$$f(0.625) = 0.2598$$

$$C = \frac{0.5 + 0.625}{2} = 0.5625$$

$$f(0.5625) \rightarrow -0.1618$$

$$f(0.5625) \cdot f(0.625) < 0$$

$$\text{Hata} \rightarrow \frac{0.625 - 0.5}{2^4} < 0.01$$

$$x_{\text{kök}} \rightarrow 0.5625$$



